UML Diagrams

mindforge

Black Sphere Industries

Generated 2026-03-23 | D2 Diagrams

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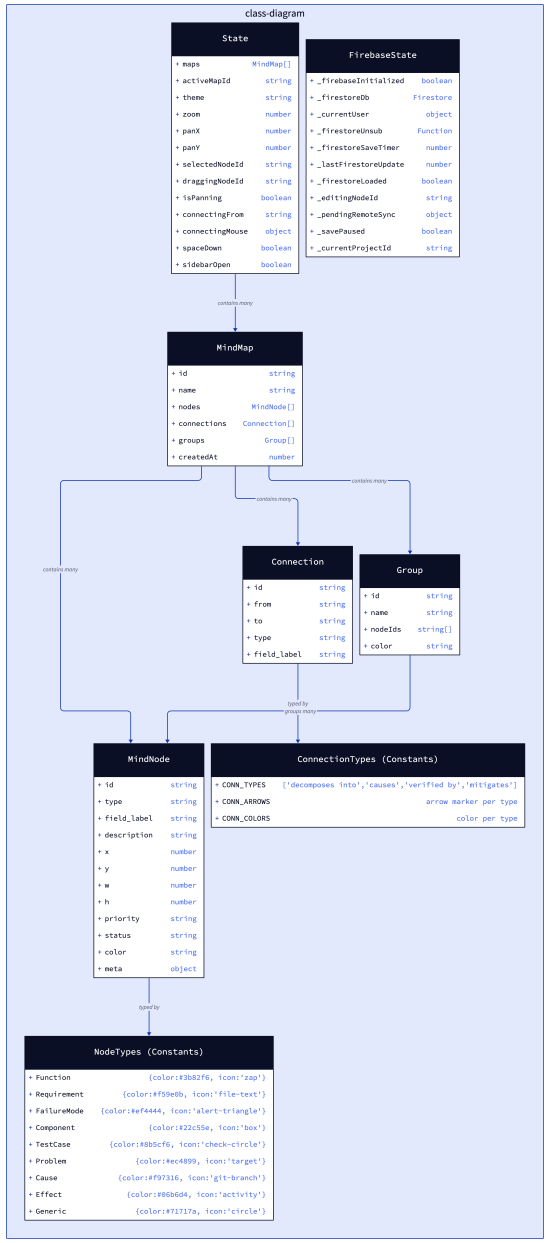
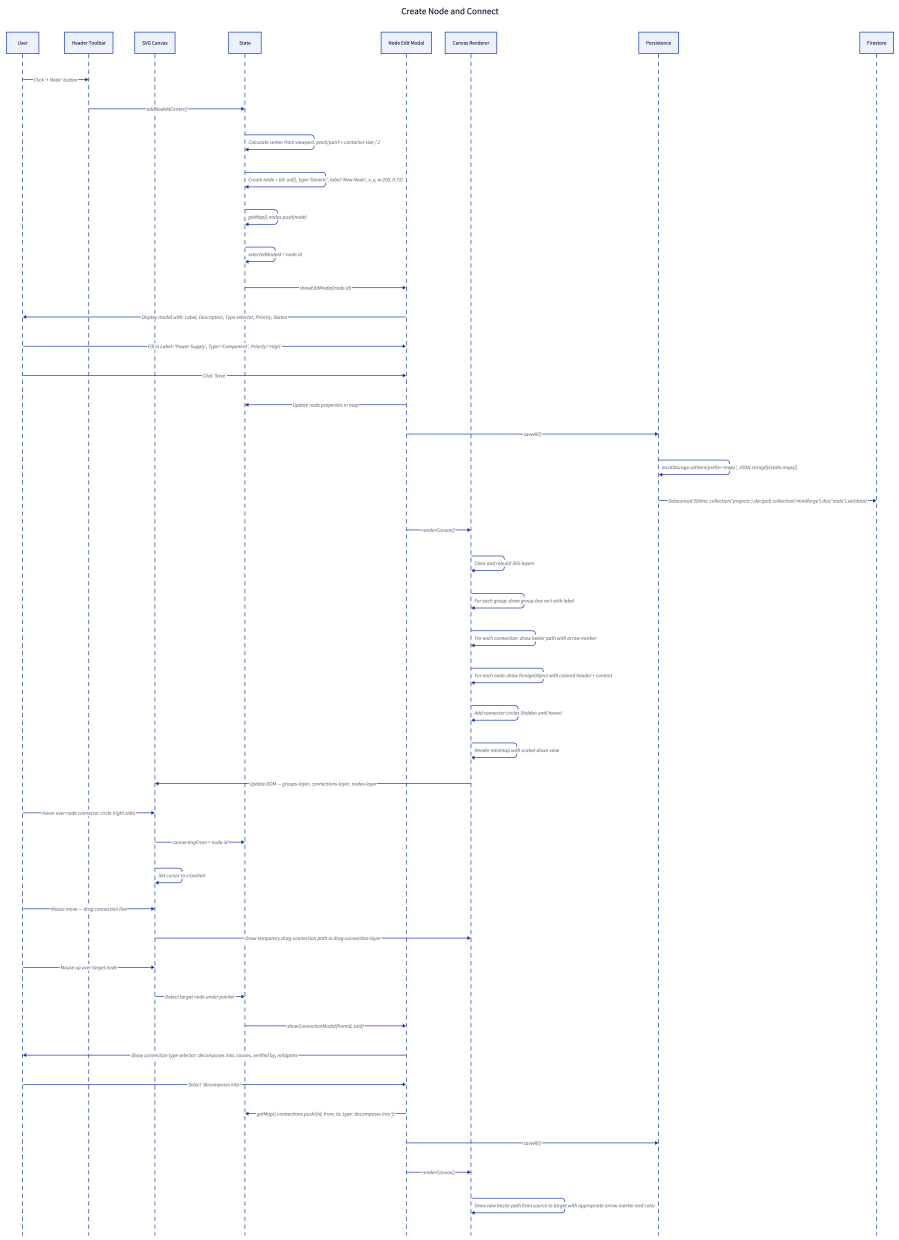
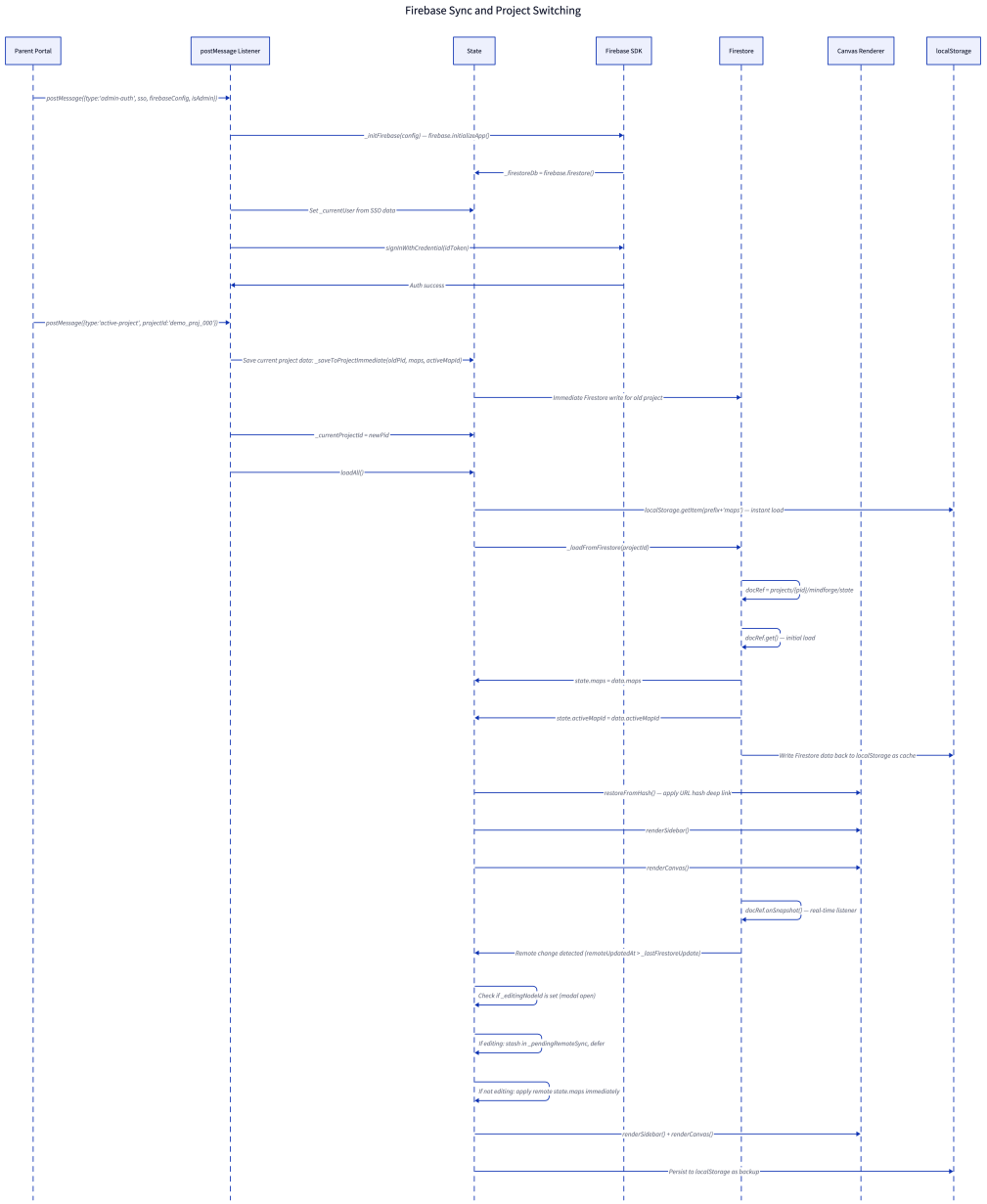
[IMG]


class-diagram
State
+ maps MindMap[]
+ activeMapId string
+ theme string
+ zoom number
+ panX number
+ panY number
+ selectedNodeId string
+ draggingNodeId string
+ isPanning boolean
+ connectingFrom string
+ connectingMouse object
+ spaceDown boolean
+ sidebarOpen boolean
FirebaseState
+ _firebaseInitialized boolean
+ _firestoreDb Firestore
+ _currentUser object
+ _firestoreUnsub Function
+ _firestoreSaveTimer number
+ _lastFirestoreUpdate number
+ _firestoreLoaded boolean
+ _editingNodeId string
+ _pendingRemoteSync object
+ _savePaused boolean
+ _currentProjectId string
contains many
MindMap
+ id string
+ name string
+ nodes MindNode[]
+ connections Connection[]
+ groups Group[]
+ createdAt number
contains many
contains many
contains many
Connection
+ id string
+ from string
+ to string
+ type string
+ field_label string
Group
+ id string
+ name string
+ nodeIds string[]
+ color string
typed by
groups many
MindNode
+ id string
+ type string
+ field_label string
+ description string
+ x number
+ y number
+ w number
+ h number
+ priority string
+ status string
+ color string
+ meta object
ConnectionTypes (Constants)
+ CONN_TYPES ['decomposes into','causes','verified by','mitigates']
+ CONN_ARROWS arrow marker per type
+ CONN_COLORS color per type
typed by
NodeTypes (Constants)
+ Function {color:#3b82f6, icon:'zap'}
+ Requirement {color:#f59e0b, icon:'file-text'}
+ FailureMode {color:#ef4444, icon:'alert-triangle'}
+ Component {color:#22c55e, icon:'box'}
+ TestCase {color:#8b5cf6, icon:'check-circle'}
+ Problem {color:#ec4899, icon:'target'}
+ Cause {color:#f97316, icon:'git-branch'}
+ Effect {color:#06b6d4, icon:'activity'}
+ Generic {color:#71717a, icon:'circle'}

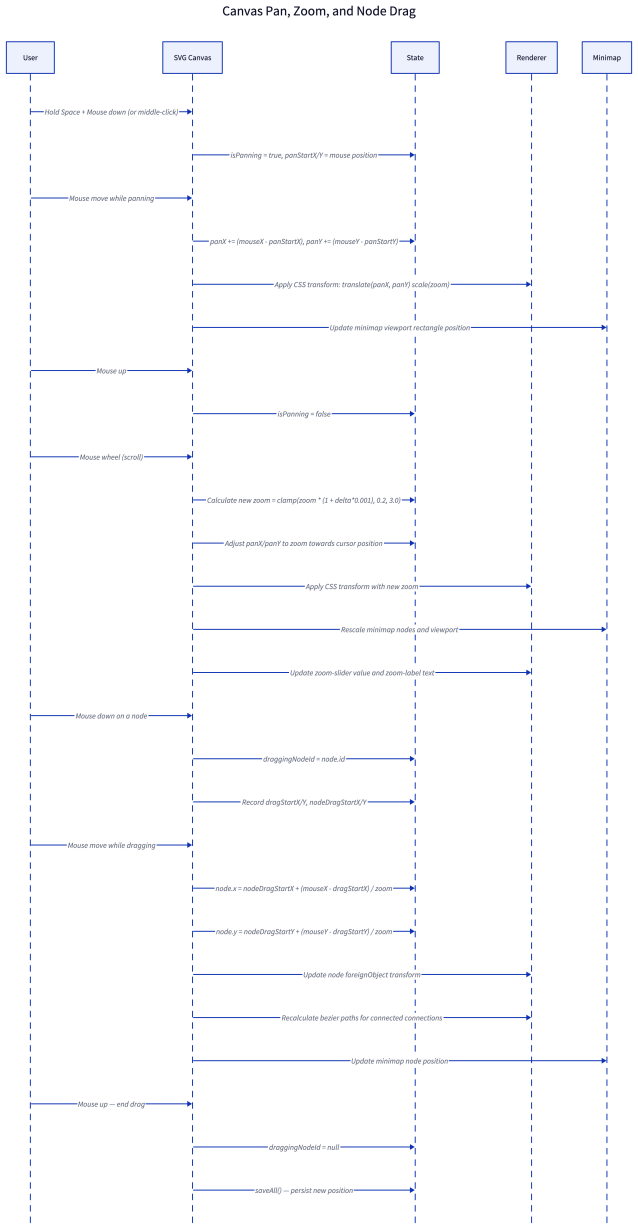
Create Node and Connect



Firebase Sync and Project Switching



Canvas Pan, Zoom, and Node Drag



```
architecture: {
  browser: Browser {
    shape: cloud
    style.stroke: "#f59e0b"

    mindforge: MindForge App {
      ui-layer: UI Layer {
        sidebar: Sidebar {
          map-list: Map List
          import-export: Import / Export Buttons
        }
        header: Header {
          map-name: Map Name Display
          zoom-controls: Zoom Slider + Fit
          add-node-btn: "+ Node" Button
          theme-toggle: Theme Toggle
          user-info: User Info (SSO)
        }
        canvas-area: Canvas Area {
          svg-canvas: SVG Canvas {
            grid-layer: Grid Pattern Layer
            groups-layer: Groups Layer
            connections-layer: Connections Layer (bezier paths)
            nodes-layer: Nodes Layer (foreignObject)
            drag-conn-layer: Drag Connection Layer
          }
          minimap: Minimap (scaled SVG)
        }
        context-menu: Context Menu {
          edit: Edit Node
          change-type: Change Type Submenu
          push-to-app: Push to App Submenu
          delete: Delete Node
        }
        modal: Modal (Edit Node / New Map)
      }

      state-layer: State Layer {
        state-obj: State Object (maps, zoom, pan, selection)
        node-types: Node Types Constants
        conn-types: Connection Types Constants
      }

      persistence: Persistence Layer {
        localstorage: localStorage (instant backup)
        firestore-sync: Firestore Sync (debounced)
      }

      theme-bridge: BSI Theme Bridge
    }
  }

  firebase-cloud: Firebase Cloud {
    shape: cloud
    style.stroke: "#ef4444"
    auth: Firebase Auth (SSO)
    firestore: Firestore {
```

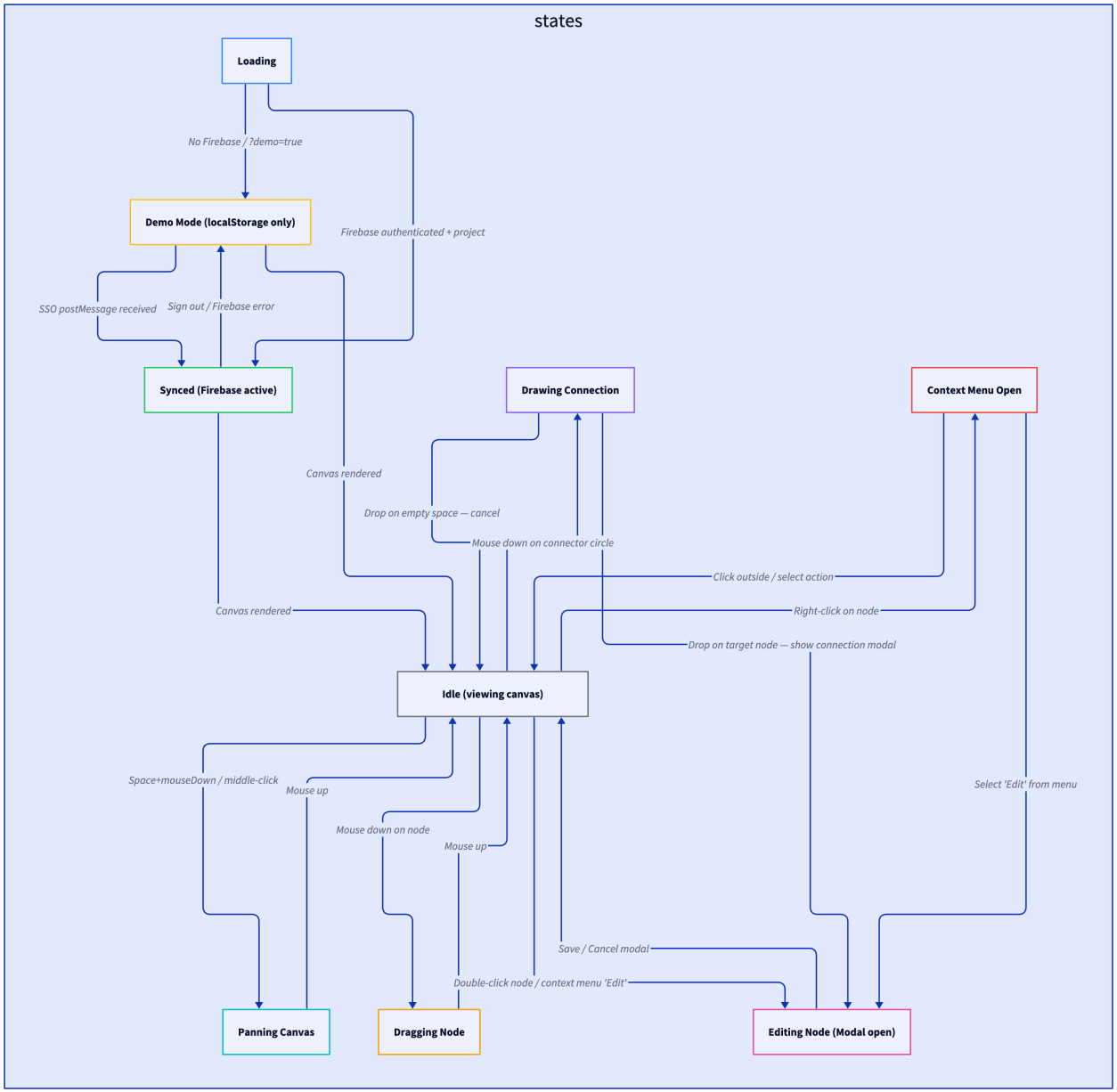
```
      shape: cylinder
      projects-col: "projects/{pid}/mindforge/state"
    }
  }

  cdn: CDN Libraries {
    shape: cloud
    style.stroke: "#3b82f6"
    tailwind: Tailwind CSS
    lucide: Lucide Icons
    jszip: JSZip
    firebase-compat: Firebase Compat SDK
  }

  portal: BSI Portal {
    shape: hexagon
    sso: SSO Auth
    project-context: Active Project Context
    theme: Theme Controller
  }

  cross-app: Other BSI Apps {
    shape: hexagon
```

... (90 lines total)


states
Loading
No Firebase / ?demo=true
Demo Mode (localStorage only)
Firebase authenticated + project
SSO postMessage received
Sign out / Firebase error
Synced (Firebase active)
Drawing Connection
Context Menu Open
Canvas rendered
Drop on empty space — cancel
Mouse down on connector circle
Click outside / select action
Canvas rendered
Right-click on node
Drop on target node — show connection modal
Idle (viewing canvas)
Space+mouseDown / middle-click
Mouse up
Select 'Edit' from menu
Mouse down on node
Mouse up
Save / Cancel modal
Double-click node / context menu 'Edit'
Panning Canvas
Dragging Node
Editing Node (Modal open)